WORKED SOLUTION

STEP III 2012, Question 10

Mechanics • Circular Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

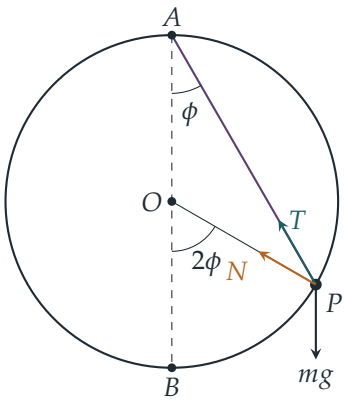
step.maths.org/questions/12-s3-q10

The key idea

Everything is geometry of the isosceles triangle formed by the centre of the hoop, the top of the hoop and the ring. Once we see that the string has length $2a\cos\phi$ and that the ring's radius makes angle 2ϕ with the downward vertical, the problem is one energy equation (for λ and the speed) and one radial equation of motion (for the reaction).

Solution

We model the ring as a particle moving on a smooth circle, with a light elastic string obeying Hooke's law. Let O be the centre of the hoop, A its highest point, B its lowest point and P the ring, and write $c = \cos\phi$.



Geometry

Triangle OAP has $OA = OP = a$, so it is isosceles and the angles at A and P are both ϕ. Hence

$$AP = 2a\cos\phi, \qquad \angle AOP = \pi - 2\phi, \qquad \angle BOP = 2\phi.$$

So the depth of P below O is $a\cos 2\phi$, and its height above B is $a(1-\cos 2\phi) = 2a\sin^2\phi$. The extension of the string (while taut) is $x = 2a\cos\phi - a = a(2c-1)$, and the string goes slack when $2c-1=0$, that is, at $\phi = \frac{\pi}{3}$.

The modulus of elasticity

The hoop is smooth and the reaction is perpendicular to the motion, so mechanical energy is conserved. At B ($\phi = 0$) the ring is at rest with extension a, so the elastic energy is $\lambda a^2/(2a) = \frac{1}{2}\lambda a$. At $\phi = \frac{\pi}{3}$ it is at rest again, the string has just become slack (no elastic energy), and it has risen $2a\sin^2\frac{\pi}{3} = \frac{3}{2}a$. Therefore

$$\tfrac{1}{2}\lambda a = mg\cdot\tfrac{3}{2}a \quad\Longrightarrow\quad \lambda = 3mg.$$

The speed

At a general angle ϕ with the string taut, energy conservation from B gives

$$\begin{aligned}\tfrac{1}{2}mv^2 &= \tfrac{1}{2}\lambda a - \frac{\lambda a^2(2c-1)^2}{2a} - 2mga\sin^2\phi\\ &= \tfrac{3}{2}mga[1-(2c-1)^2] - 2mga(1-c^2)\\ &= \tfrac{3}{2}mga(4c-4c^2) - 2mga + 2mgac^2\\ &= mga(-4c^2+6c-2),\end{aligned}$$

so

$$\frac{mv^2}{a} = mg(-8c^2+12c-4) = -4mg(2c-1)(c-1).$$

This vanishes at $c=1$ and $c=\frac{1}{2}$ and is positive in between, consistent with the ring moving between $\phi = -\frac{\pi}{3}$ and $\phi = \frac{\pi}{3}$: after coming to rest it slides back through B to the symmetric point on the other side, and so on. Throughout, $\frac{1}{2}\le c\le 1$ and the string is taut (just slack at the extreme points), with tension

$$T = \frac{\lambda x}{a} = 3mg(2c-1).$$

The reaction

Let N be the component of the reaction towards O, as drawn (it may turn out negative). We resolve towards the centre along PO:

- the tension acts along PA, which makes angle ϕ with PO (the base angle of the isosceles triangle), giving $T\cos\phi$ towards O;
- OP makes angle 2ϕ with the downward vertical, so the weight has component $mg\cos 2\phi$ directed *away* from O.

Ask yourself Why is the weight's component outward rather than inward here?

For $\phi < \frac{\pi}{4}$ the ring is below O and gravity pulls it away from the centre; the single expression $-mg\cos 2\phi$ (towards O) covers both cases, because $\cos 2\phi$ changes sign when P passes the level of O.

Newton's second law towards the centre gives

$$N + T\cos\phi - mg\cos 2\phi = \frac{mv^2}{a}.$$

Substituting T, mv^2/a and $\cos 2\phi = 2c^2 - 1$:

$$\begin{aligned} N &= mg(-8c^2 + 12c - 4) - 3mg(2c-1)c + mg(2c^2 - 1) \\ &= mg(-8c^2 + 12c - 4 - 6c^2 + 3c + 2c^2 - 1) \\ &= -mg(12c^2 - 15c + 5). \end{aligned}$$

Hence the magnitude of the reaction is

$$R = (12\cos^2\phi - 15\cos\phi + 5)\,mg,$$

as required, provided the bracket is never negative, which we now show. (The minus sign tells us the hoop pushes the ring *outwards*, away from O, throughout.)

R is never zero

The quadratic $12c^2 - 15c + 5$ has discriminant $15^2 - 4\cdot 12\cdot 5 = 225 - 240 = -15 < 0$ and positive leading coefficient, so it is strictly positive for every real c. Completing the square,

$$12c^2 - 15c + 5 = 12\left(c - \tfrac{5}{8}\right)^2 + \tfrac{5}{16} \geq \tfrac{5}{16},$$

so $R \geq \frac{5}{16}mg > 0$ throughout the motion, with the minimum when $\cos\phi = \frac{5}{8}$ (which lies in $[\frac{1}{2}, 1]$, so it is actually attained).

Checks

- At B ($c = 1$): the ring is at rest, $T = 3mg$ upwards, weight mg downwards, so the hoop must push down with $2mg$. The formula gives $(12 - 15 + 5)mg = 2mg$.
- At the slack point ($c = \frac{1}{2}$): no tension and no speed, P is $\frac{a}{2}$ above O, so the radial reaction balances the inward weight component $mg\cos\frac{\pi}{3} = \frac{1}{2}mg$. The formula gives $(3 - 7.5 + 5)mg = \frac{1}{2}mg$.
- We integrated the equation of motion numerically (RK4, starting just off B): the ring stops at angle $2.094 \approx \frac{2\pi}{3}$ from B, the computed reaction matches the formula to within 10^{-7}, and its minimum is $0.3125\,mg = \frac{5}{16}mg$.

Common slips

- Taking the ring's angular position to be ϕ rather than 2ϕ: the angle at the circumference is half the angle at the centre.

- Using the full string length $2a\cos\phi$ instead of the extension $2a\cos\phi - a$ in Hooke's law or the elastic energy.
- Assuming the reaction points inwards and then being puzzled by a negative answer; the question asks only for its magnitude.

Beyond STEP

The ring has one degree of freedom, and the cleanest way to handle such systems is through an energy (or Lagrangian) formulation in a single coordinate, with constraint forces like R recovered afterwards from Newton's law. That split, between equations of motion from energy and constraint forces from resolving, is exactly how first-year mechanics courses treat beads on wires. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers circular motion with constraints and energy methods in this spirit.

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